Hybrid-Level Instruction Injection for Video Token Compression in Multi-modal Large Language Models

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Abstract

*Recent Multi-modal Large Language Models (MLLMs) have been challenged by the computational overhead resulting from massive video frames, often alleviated through compression strategies. However, the visual content is not equally contributed to user instructions, existing strategies (e.g., average pool) inevitably lead to the loss of potentially useful information. To tackle this, we propose the **H**ybrid-level Instruction **I**njection Strategy for **C**onditional Token **Com**pression in MLLMs (HICom), utilizing the instruction as a condition to guide the compression from both local and global levels. This encourages the compression to retain the maximum amount of user-focused information while reducing visual tokens to minimize computational burden. Specifically, the instruction condition is injected into the grouped visual tokens at the local level and the learnable tokens at the global level, and we conduct the attention mechanism to complete the conditional compression. From the hybrid-level compression, the instruction-relevant visual parts are highlighted while the temporal-spatial structure is also preserved for easier understanding of LLMs. To further unleash the potential of HICom, we introduce a new conditional pre-training stage with our proposed dataset HICom-248K. Experiments show that our HICom can obtain distinguished video understanding ability with fewer tokens, increasing the performance by 2.43% average on three multiple-choice QA benchmarks and saving 78.8% tokens compared with the SOTA method. The code is available at* https://github.com/lntzm/HICom.

1. Introduction

Multi-modal Large Language Models [2, 14, 22, 31, 32, 74] (MLLMs) have gained significant improvements in multi-modal understanding by integrating visual information into the LLMs, beating various expert methods on downstream tasks [12, 13, 25, 26, 47, 61, 65, 70]. While initially focused on image understanding, more research [30, 34, 36, 40, 56, 57] has shifted towards challenging video tasks. Most MLLMs handle videos by treating them as sequences of images, sampling frames, and concatenating visual tokens from these frames [6, 20, 42, 52, 66].

Compared to images, videos comprise multiple frames, resulting in more visual tokens and thus higher computational costs. To make the computation affordable, early methods [23, 30, 34] often employ sparse temporal sampling strategies, leading to significant loss of temporal information. Consequently, current MLLMs focus on compress-

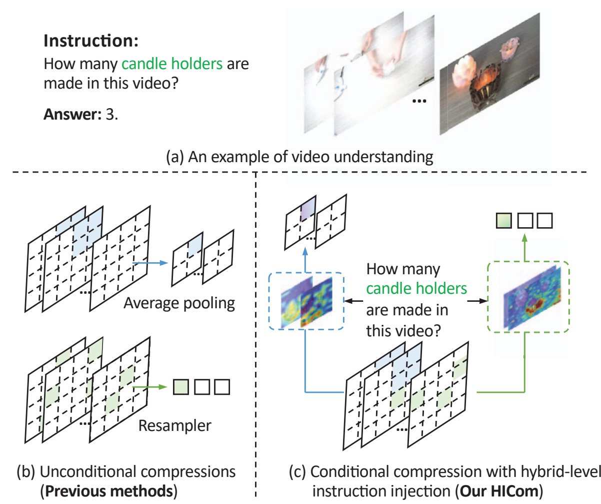

Figure 1. An example of the video understanding task, and the comparison between the unconditional compression and our proposed conditional compression with hybrid-level instruction injection. We inject instruction at both local and global levels, guiding the compression to retain the maximum amount of user-focused information and minimize the computational burden.

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ing visual tokens to achieve a trade-off between computational costs and video understanding ability. Mainstream MLLMs [20, 59, 71] use a spatial pooling strategy within each frame, leveraging capabilities gained from image data. Some approaches [19, 43, 67] use Q-Former [22] or Resampler [1] to compress the spatial-temporal information into a fixed number of visual tokens. Other methods employ convolution [7], clustering [17], or memory mechanisms [16, 46, 68] to compress visual tokens in both spatial and temporal dimensions. However, these compression methods are unconditional, lacking explicit guidance, as illustrated in Fig. 1 (b), making it challenging to achieve high compression ratios with minimal information loss. Consequently, user-relevant visual content may be overlooked during such unconditional compression.

Therefore, we focus on introducing the instruction information as a condition to perform targeted compression, as it is rarely explored in previous work. As shown in Fig. 1 (c), conditional compression uses explicit instructional guidance to select user-focused information for retention, thereby enhancing efficiency. To thoroughly explore the conditional compression, we propose **H**ybrid-level Instruction **I**njection Strategy for **C**onditional Token **Com**pression in MLLMs (HICom). Observing how humans process complex visual information, they usually adopt a coarse-to-fine approach, first considering the relative parts in the global coarse impression and then seeking detailed information locally in the video. Motivated by this human perception process, HICom conducts the compression at both local and global levels, retaining the maximum amount of instruction-relevant information while preserving the temporal-spatial structure with fewer tokens, thus achieving a better balance between the computational burden and video comprehension.

Specifically, at the local level, we divide the sampled frame features into several groups and compress the tokens of each group into one token based on the attention with the injected instruction condition. Different from Q-Former [22] and Resampler [1], the grouped local attention can explicitly maintain the spatial-temporal structural characteristics of the video while highlighting the relevant parts within each group. At the global level, we inject the instruction condition into a small number of learnable tokens and then conduct the attention with the flattened frame features without grouping. The global compression focuses more on searching instruction-relevant information in the whole video and thus can be expressed by a small number of tokens as an auxiliary. To further unleash the potential of HICom, we introduce a new conditional pre-training stage between the alignment and instruction tuning and construct a new dataset HICom-248K of 248K video clips with high-quality instruction-followed descriptions. The proposed new stage continues to pre-train the parameters of condition injection, further improving the performance. Experiments show that our HICom can achieve state-of-the-art performance on 5 benchmarks with significantly fewer tokens. (*e.g.*, increasing by 2.43% average on three multiple-choice QA benchmarks and saving 78.8% tokens compared with LLaVA-Video-7B [72]).

Our main contributions are listed as follows:

- We propose a conditional compression method HICom for MLLMs, achieving effective video understanding while significantly reducing the computational burden.
- We conduct hybrid-level instruction injection to achieve the conditional compression, and introduce a new conditional pre-training stage with our constructed HICom-248K dataset to further unleash the potential of the conditional compressing.
- Experiments on various benchmarks show the proposed HICom can achieve better performance with fewer visual tokens, providing valuable insights for MLLMs.

2. Related Work

Multi-modal large language models. Based on the powerful capabilities of LLMs among various linguistic tasks, many works try to extend the understanding abilities to the computer vision area. Flamingo [1] and BLIP series [8, 22] successfully explore the usage of Resampler and Q-Former to bridge the visual tokens to language models. LLaVA series [31–33] use a simple MLP as the visual projector to translate visual tokens to the LLM's embedding space, achieving huge success. Recently, more researchers have focused on extending image tasks to video tasks. The biggest challenge for video tasks is to design an effective method to achieve the trade-off between computational costs and video understanding ability. Early methods [23, 30, 34] sparsely sample frames, lossing too much temporal information. Mainstream methods [20, 59, 71] use a simple spatial pooling strategy to reduce the number of visual tokens and get a huge success. [19, 67] employ Q-Former to compress the spatial-temporal tokens into a fixed number. Chat-Univi [17] uses DPC-KNN algorithm to cluster dynamic visual tokens. VideoLLaMA2 [7] utilizes convolution both in temporal and spatial dimensions to downsample tokens. Flash-VStream [68] introduces a learnable memory mechanism to compress online video streaming. All of them try their best to observe more visual information. However, their unconditional compression without explicit guidance inherently leads to the loss of instruction-relevant information. Different from them, we conduct conditional compression by injecting instruction at hybrid levels, only emphasizing the visual parts related to the instruction and allowing the irrelevant parts to be discarded, thus getting a better representation for each question.

Text-based visual representation for video LLMs. There are some methods using instruction information during the

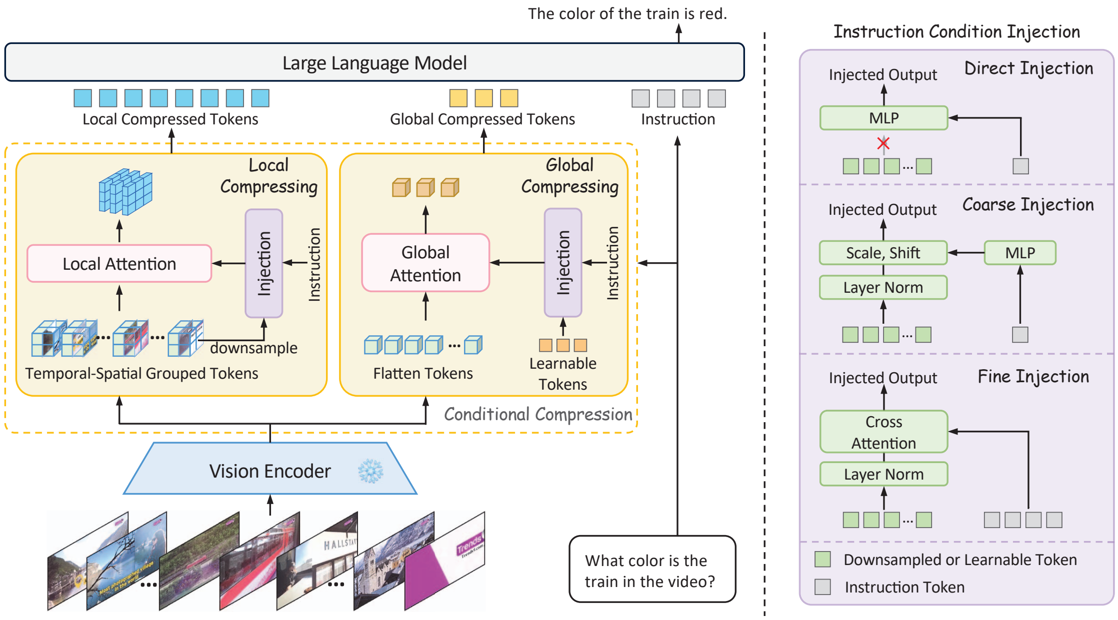

Figure 2. The framework of our proposed HICom. We propose the hybrid-level instruction injection to conditionally compress video tokens in MLLMs. We extract instruction-relevant information within each grouped sub-region at the local level, and extract it to a fixed number of tokens at the global level. The instruction condition is injected into the attention process to guide the compression.

visual encoding. [53, 63] sample the frames based on the CLIP [41] similarity or a localization model using the instruction, [29, 50] make the frame selection strategy trainable. However, these methods only focus on how to select the correct frames in a rough manner, failing to guide the token compression with the key instruction information. LLaMA-VID [28] follows a similar instruction-guided idea, as it introduces the instruction to interact with visual features. However, it roughly simplifies the interacted but uncompressed result into a single token only as auxiliary information for each frame, destroying the spatial structure and failing to discuss the guiding role of the instruction for conditional compression. Therefore, it is only designed for long videos and the performance is limited. To fully explore conditional compression, we propose to inject the instruction at hybrid levels, exploring a better way for conditional compression to alleviate the information loss while maintaining the performance.

3. Method

3.1. Instruction-Guided Conditional Compression

Fig. 2 shows the framework of our proposed HICom, which focuses on the design between the vision encoder and the LLM to conduct the conditional compression. We inject the instruction condition into the compression process at both local and global levels for better information extraction with spatial-temporal structure maintained.

Local-Level Compression. Previous works have demonstrated the importance of the spatial inductive bias when processing visual representations for image MLLMs [27, 49]. Inspired by them, we divide the tokens into several groups and perform conditional compression within each group to preserve the temporal-spatial structure of video features. Specifically, given the encoded video frame features $\boldsymbol{V} \in \mathbb{R}^{T\times H\times W\times D}$, and the downsampling ratio $(\alpha_T, \alpha_H, \alpha_W)$, we divide $\boldsymbol{V}$ into $N_T{\cdot}N_H{\cdot}N_W$ groups, where $N_i = \lceil i/\alpha_i \rceil$, $i \in \{T, H, W\}$. Therefore, there are $\alpha_T{\cdot}\alpha_H{\cdot}\alpha_W$ tokens in each group, which can be formulated as $\boldsymbol{V}^{t,h,w} \in \mathbb{R}^{\alpha_T\times\alpha_H\times\alpha_W\times D}$, where $0 \le \{t, h, w\} \le \{N_T, N_H, N_W\}$. We first pool $\boldsymbol{V}^{t,h,w}$ into one token $\boldsymbol{V}_p^{t,h,w} \in \mathbb{R}^{1\times D}$ and inject the instruction condition $\boldsymbol{C}$ (*i.e.*, the text feature of instructions) into $\boldsymbol{V}_p^{t,h,w}$. Then the local attention within each group can be calculated by:

$$\boldsymbol{Z}_l^{t,h,w} = \text{Attn}\left[\text{Inj}(\boldsymbol{V}_p^{t,h,w}, \boldsymbol{C}), \boldsymbol{V}^{t,h,w}, \boldsymbol{V}^{t,h,w}\right], \quad (1)$$

where Inj$(\cdot)$ represents the condition injection, Attn$(\cdot)$ donates the attention mechanism [9] and the inputs of Attn$(\cdot)$ are *query*, *key*, and *value*, respectively. Therefore, the visual features within each group are compressed into only one token, preserving the temporal-spatial structure while conditionally highlighting the instruction-relevant parts. Finally, we concatenate $\boldsymbol{Z}_l^{t,h,w}$ as the local compressed results $\boldsymbol{Z}_l \in \mathbb{R}^{N_T\times N_H\times N_W\times D}$, and use an MLP layer to project $\boldsymbol{Z}_l$ into the LLM's embedding space.

Global-Level Compression. Though the temporal-spatial structure can be maintained at the local level, the attention

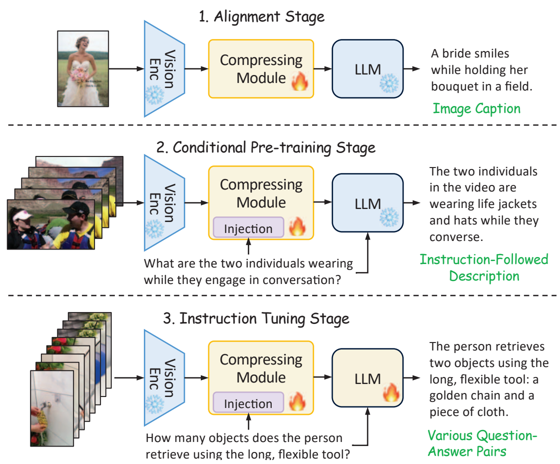

Figure 3. We introduce a new guidance pre-training stage and implement three-stage training for conditional compression.

is forced to focus on small sub-regions. However, each sub-region may contain information that is not equally valid for the question, and some sub-regions are even quite irrelevant. Therefore, we also implement conditional compression at the global level to highlight the most relevant parts within the whole video. Specifically, we initialize a small set of learnable tokens $\boldsymbol{L} \in \mathbb{R}^{N_L \times D}$, where N_L donates the number of tokens, and then inject the instruction condition into them. Instead of grouping the video frame features, we apply 3D position embedding $\mathrm{Pos}(\cdot)$ for them and directly flatten them at global-level compression. Then the global compressed tokens $\boldsymbol{Z}_g \in \mathbb{R}^{N_L \times D}$ can be calculated by:

$$\boldsymbol{Z}_g = \mathrm{Attn}\left[\mathrm{Inj}(\boldsymbol{L}, \boldsymbol{C}), \boldsymbol{V} + \mathrm{Pos}(\boldsymbol{V}), \boldsymbol{V}\right]. \tag{2}$$

An MLP layer is also utilized to project $\boldsymbol{Z}_g$ into the LLM's embedding space. And we finally concatenate $\boldsymbol{Z}_l$ and $\boldsymbol{Z}_g$ together for further understanding of LLM.

Instruction Condition Injection. In order to fulfill the guiding role of the instruction condition, we explore three different types of injection modules, and define them direct injection, coarse injection, and fine injection, respectively. Note the text encoder can obtain both pooled tokens $\boldsymbol{C}_p \in \mathbb{R}^{1 \times D}$ (*i.e.*, global text embedding) and fine-grained tokens $\boldsymbol{C}_f \in \mathbb{R}^{L \times D}$ (*i.e.*, token-level text embedding). Given the pooled token $\boldsymbol{C}_p$, the direct injection uses an MLP to translate the single token into the injected output without any interaction with $\boldsymbol{V}_p^{t,h,w}$ or $\boldsymbol{L}$. We employ coarse injection by the adaptive layer norm [39]. Specifically, we use an MLP to regress the scale and shift from $\boldsymbol{C}_p$, then add to the visual input $\boldsymbol{V}_p^{t,h,w}$ or learnable tokens $\boldsymbol{L}$, which we represent them as $\boldsymbol{A}$, after the layer norm. The process can be formulated as:

$$\mathrm{Inj}(\boldsymbol{A}, \boldsymbol{C}) = \mathrm{LN}(\boldsymbol{A}) \cdot \mathrm{scale}(\boldsymbol{C}_p) + \mathrm{shift}(\boldsymbol{C}_p), \tag{3}$$

Table 1. The statistical information of our constructed HICom-248K dataset.

# Videos	Sources	Avg Length	# QA Pairs
248K	Panda70M: 238K Ego4D: 10K	25.67s	739K

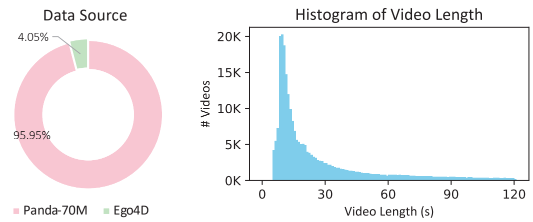

Figure 4. The visualization of data source (left) and video length (right) of our constructed HICom-248K dataset.

where $\mathrm{LN}(\cdot)$ is the layer norm. Given the fine-grained tokens $\boldsymbol{C}_f$ as condition $\boldsymbol{C}$, we conduct the fine injection via the cross attention, which can be formulated as:

$$\mathrm{Inj}(\boldsymbol{A}, \boldsymbol{C}) = \mathrm{Attn}\left(\mathrm{LN}(\boldsymbol{A}), \boldsymbol{C}_f, \boldsymbol{C}_f\right). \tag{4}$$

As the direct injection directly translates the condition to the *query* of the attention in Eq. (1) and Eq. (2), the token length limits that it is only suitable for local compression as the local branch compresses each group into only one token. On the contrary, the coarse and fine injections are more flexible to fit various situations as the condition is added into $\boldsymbol{A}$. We conduct the experiments on the three types of injection and finally choose direct injection for local compression and coarse injection for global compression.

Conditional Pre-training Stage. The existing mainstream methods follow the pipeline of alignment first and then instruction tuning [7, 31, 32, 68], and previous text-based methods [28, 53] also follow this pipeline. However, the image-caption pairs cannot provide valid instruction information in the alignment stage, direct instruction tuning with modules that are not adequately aligned will confuse the guiding process, thus harming the performance. Therefore, we propose a new conditional pre-training stage between the alignment and the instruction tuning to make the training process easier, as shown in Fig. 3. In this stage, we use our constructed instruction-followed descriptions to pre-train the compression module with condition injection, fulfilling the guiding role of the instruction.

3.2. Dataset Construction

To pre-train the condition injection in the compression module, we construct a new instruction-followed description dataset HICom-248K. Different from common instruction tuning datasets, HICom-248K focuses on providing only one data type, *i.e.*, the description of the visual content re-

Table 2. Performance comparison between our HICom and other SOTA methods on multiple-choice QA video benchmarks. † means we use the result reproduced by VideoLLaMA2 [7], $*$ indicates we reproduce the results ourselves using the official checkpoint and inference code provided by authors. § donates we inference with a new length of frames trained by sampling 32 frames.

Methods	**LLM Size**	**# Frames**	**# Tokens**	**VideoMME w/o sub.**				**VideoMME w/ sub.**				**MV-Bench**	**Ego-Schema**
				short	**mid**	**long**	**all**	**short**	**mid**	**long**	**all**		
Video-LLaVA [30]	7B	8	2048	45.3	38.0	36.2	39.9	46.1	40.7	38.1	41.6	43.0	-
VideoChat2-Mistral [24]	7B	16	1536	48.3	36.3	35.0	39.5	52.8	39.4	39.2	43.8	60.4	-
LLaMA-VID [28]	7B	1fps	2tps	-	-	-	25.9†	-	-	-	-	41.9†	38.5†
Chat-Univi-1.5 [17]	7B	64	448	45.7	40.3	35.8	40.6	51.2	44.6	41.8	45.9	45.9	-
PLLaVA [59]	7B	16	2304	-	-	-	-	-	-	-	-	46.6	-
LLaVA-Next-Video [71]	7B	32	4608	49.9*	41.4*	37.0*	42.8*	-	-	-	-	46.5†	43.9†
LongVA [69]	7B	128	18432	61.1	50.4	46.2	52.6	61.6	53.6	47.6	54.3	-	-
VideoLLaMA2 [7]	7B	16	1152	56.0	45.4	42.1	47.9	-	-	-	-	54.6	51.7
Tarsier [51]	7B	16	2304	-	-	-	-	-	-	-	-	62.6	56.0
VITA [11]	8×7B	16	4096	65.9	52.9	48.6	55.8	70.4	56.2	50.9	59.2	-	-
LLaVA-OneVision [20]	7B	32	6272	70.1*	56.4*	48.9*	58.5*	75.8*	58.4*	51.6*	61.9*	56.7	60.1
LLaVA-Video [72]	7B	32	6272	**73.9***	57.3*	50.4*	**60.6***	**76.6***	60.3*	51.7*	62.9*	58.6	57.3
HICom (Ours)	1.5B	32	680	61.7	51.3	43.9	52.3	65.1	54.7	47.3	55.7	56.4	50.1
HICom (Ours)	7B	32	680	65.7	57.1	51.4	58.1	68.4	58.9	53.1	60.1	64.1	60.5
HICom (Ours)§	7B	64	1328	67.6	58.0	51.2	58.9	71.8	63.7	54.5	63.3	**64.7**	62.2
HICom (Ours)§	7B	128	2624	69.0	**60.2**	**51.5**	60.3	72.7	**64.6**	**57.3**	**64.8**	64.0	**62.3**

lated to the instruction, and does not include other rich types of instruction data (*e.g.*, reasoning, counting, *etc*).

Data Collection. To ensure the diversity and quality of our videos, we collect videos from public datasets Panda-70M [5] and Ego4D [15]. Panda-70M contains various open-domain videos from YouTube, and Ego4D collects many high-resolution ego-centric human activity videos. Notably, we use the original untrimmed videos to cut clips ourselves since the provided split clips of both datasets are too short, greatly decreasing the content complexity. To balance the number of different types of videos, we pre-define 29 categories [10, 72] using natural language (*e.g. A video about cooking activity.*) and use InternVideo2 [54] to extract both the video features and the pre-defined sentence features. Based on the similarities between them, we select 1,500 videos for each category and then randomly select additional 10,000 videos from the others to ensure diversity.

Video Processing. The untrimmed videos are too long for our pre-training, and also severely affect the quality of captions and annotations by existing SOTA models. Therefore, we use PySceneDetect to split each video into shorter ones with a much looser threshold than Panda-70M. As the split clips can sometimes be redundant with repeat semantics, we further extract keyframes for each video and only select the split clips containing the keyframes as our final results. Specifically, inspired by ShareGPT4Video [4], we calculate the CLIP score for densely sampled frames within a video, and select the less similar frames as keyframes. Finally, we further filter out the video clips shorter than 5 seconds and longer than 120 seconds.

Instruction-Followed Description Generation. For the processed video clips, we use Qwen2-VL-72B-Instruct [52] to systematically generate the instruction-followed description with the CoT technique [55]. Specifically, we first generate the detailed caption of each input video, then take both the video content and caption as input to generate the three instruction-answer pairs for each video. Different from the normal instruction datasets, we require Qwen2-VL-72B-Instruct to meet the following two rules: 1) The instructions should refer to the specific visual information, such as the people or the objects in the video; 2) The instructions should lead to a descriptive response and the answers should describe the object mentioned in the instruction in detail. Finally, we filter the generated results by discarding answers containing invalid information, such as the phrase ”not provide”, or ”not describe”.

Dataset Statistics. Tab. 1 and Fig. 4 shows some statistics about our constructed dataset. We collect 248K video clips in total from Panda-70M and Ego4D, the average of the video lengths is 25.67 seconds and the distribution is shown in Fig. 4. With the help of Qwen2-VL-72B-Instruct, we generate 739K instruction-followed descriptions, providing sufficient data for our guidance pre-training.

4. Experiments

4.1. Experimental Setup

Implementation Details. We use SigLIP [64] (so400m-patch14-384) as our vision encoder and text encoder, choose Qwen2.5 series [48] as our LLMs, and randomly initialize the compressor. The vision encoder keeps frozen at all stages, the LLM is frozen at the two pre-train stages, and is fine-tuned at the instruction tuning stage. We follow LLaVA-OneVision [20] to choose our training configurations, the global batch size is set to 512 at the alignment stage, 256 at the conditional pre-train and instruction tun-

Table 3. Performance of our HICom and other SOTA methods on open-ended QA video benchmarks. All the compared models are 7B. †, ∗, and § keep the same meaning with Tab. 2. *# F.*, *# T.* are short for # Frames, # Tokens, ANet is short for ActivityNet, LLaVA-NV, LLaVA-OV are short for LLaVA-Next-Video, and LLaVA-OneVision. The format we report ANet is Acc/Score.

Methods	# F.	# T.	ANet	VideoChatGPT Bench					
				CI	DO	CU	TU	CO	Avg
Video-LLaVA [30]	8	2048	45.3/3.3	-	-	-	-	-	-
VideoChat2 [24]	16	1536	49.1/3.3	3.02	2.88	3.51	2.66	2.81	2.98
Chat-Univi [17]	64	448	45.8/3.2	2.89	2.91	3.46	2.89	2.81	2.99
LLaMA-VID [28]	1fps	2tps	47.4/3.3	2.96	3.00	3.53	2.46	2.51	2.89
LongVLM [58]	100	305	47.6/3.3	2.76	2.86	3.34	2.39	3.11	2.89
ST-LLM [35]	16	512	50.9/3.3	3.23	3.05	3.74	**2.93**	2.81	3.15
PLLaVA [59]	16	2304	56.3/3.5	3.21	2.86	3.62	2.33	2.93	2.99
LLaVA-NV [71]	32	4608	53.5/3.2	3.39	**3.29**	**3.92**	2.6	3.12	**3.26**
LongVA [69]	128	18432	-/2.8	3.05	3.09	3.77	2.44	**3.64**	3.20
VideoLLaMA2 [7]	16	1152	50.2/3.3	3.16	3.08	3.69	2.56	3.14	3.13
Tarsier [51]	16	2304	**59.5**/3.6	-	-	-	-	-	-
SF-LLaVA [60]	50	3680	56.3/3.4	3.09	2.70	3.57	2.52	3.35	3.04
LLaVA-OV [20]	32	6272	56.6/3.6*	**3.45***	3.00*	3.71*	2.68*	3.14*	3.20*
HICom(Ours)-1.5B	32	680	53.0/3.5	3.09	2.67	3.40	2.41	2.98	2.91
HICom(Ours)-7B	32	680	58.3/**3.7**	3.29	2.85	3.59	2.67	3.22	3.12
HICom(Ours)-7B§	64	1328	59.4/**3.7**	3.32	2.92	3.65	2.74	3.35	3.20
HICom(Ours)-7B§	128	2624	**59.5/3.7**	3.33	2.86	3.65	2.74	3.32	3.18

ing stage. We use the learning rate of 1e-3 at the alignment stage, 1e-5 at the instruction tuning stage. At our proposed conditional pre-training stage, we use 1e-3 for the condition injection sub-module and 1e-4 for other parameters in the compressing module. We train 1 epoch for all stages. More implementation details can be found in Appendix A.

Datasets. We use LLaVA-558K [32] image-caption pairs for our alignment stage, the constructed 248K HICom-Pretrain instruction-followed descriptions for conditional pre-train stage. Inspired by [51, 72], we collect 2.68M video instruction data for instruction tuning, including 1.6M from LLaVA-Video [72], 292K from VideoChat2-IT [23] subset, 255K from M4-Instruct-Data [21], 11K from Charades [45], 114K from NTU RGB+D [44], 122K from TVQA [18], and 290K from MiT [38]. We evaluate our model on 5 video benchmarks, including 3 multiple-choice benchmarks VideoMME [10], MVBench [24], EgoSchema [37], and 2 open-ended benchmarks ActivityNet [3], VideoChatGPT Bench [36].

4.2. Main Results

Multiple-choice benchmarks. In Tab. 2, we compare the performance of our HICom with different SOTA models on three multiple-choice video benchmarks. We sample 32 frames with 680 tokens to train all our models and sample different frames for inference. Our HICom-7B with 2624 tokens inference achieves the best performance on all three benchmarks. Compared to LLaVA-Video with 6272 tokens, our HICom with only 1328 tokens can obtain better performance, increasing the performance by 2.43% average and

Table 4. The ablation study on our proposed HICom, $\alpha_{(T,H,W)}$ donates the downsampling ratio of each dimension, *# Tok.* donates the number of tokens input to LLMs. 8f means we sample 8 frames for training. The conditional compression combining both local and global achieves the best.

Methods	$\alpha_{(T,H,W)}$	# Tok.	VideoMME w/o sub.				MV-Bench	Ego-Schema
			short	mid	long	all		
Unconditional Compression								
avg pool	(1,3,3)	2592	39.3	34.9	33.0	35.7	44.8	43.2
8f, avg pool	(1,3,3)	648	36.3	34.4	32.0	34.3	43.6	42.7
avg pool	(4,3,3)	648	36.3	33.3	32.2	34.0	43.7	41.9
local	(4,3,3)	648	36.7	34.4	32.0	34.4	43.7	**42.7**
global	None	32	30.0	31.6	28.9	30.1	34.6	33.6
local+global	(4,3,3)	680	**37.1**	**34.8**	**32.3**	**34.7**	**44.1**	42.4
Conditional Compression								
local	(4,3,3)	648	38.8	36.1	33.1	36.0	44.0	43.2
global	None	32	30.8	30.3	29.8	30.3	35.5	34.0
local+global	(4,3,3)	680	**39.0**	**36.7**	**34.2**	**36.6**	**45.0**	**43.5**

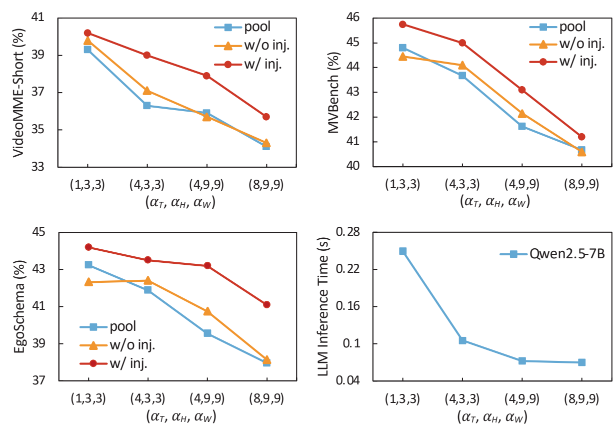

Figure 5. The ablation study on different compressing ratios. The figures show the performance on VideoMME-Short (upper left), MVBench (upper right), EgoSchema (lower left), and the inference time of 7B LLM (lower right).

saving 78.8% tokens, HICom with 2624 tokens gains 3% averagely when saving 58.2% tokens, significantly lowering the computational burden. Notably, LLaVA-Video uses numerous additional image data, and it also unfreezes the vision encoder during training, which can both increase performance. It is also noteworthy that though our HICom performs a little worse than LLaVA-Video on VideoMME short videos (less than 2 minutes), we beat LLaVA-Video on both VideoMME medium (4-15 minutes) and long videos (30-60 minutes). We argue that conditional compression is more helpful for long videos. For short videos, the sampled frames are dense enough for unconditional compression, baselines with enough information would naturally perform powerfully. But for long videos, the sampled frames are more sparse with less redundant, our HICom can preserve

Table 5. The ablation study on the conditional pre-training stage with the constructed HICom-248K dataset.

Methods	Conditional Pre-trained	VideoMME w/o subs.				MV-Bench	Ego-Schema
		short	mid	long	all		
avg pool	✗	36.6	32.4	33.2	34.1	42.7	40.6
	✓	36.3	33.3	32.2	34.0	43.7	41.9
w/o inj.	✗	39.4	33.7	33.1	35.4	42.8	41.64
	✓	37.1	34.8	32.3	34.7	44.1	42.4
w/ inj.	✗	38.8	35.1	34.1	36.0	44.0	41.6
	✓	39.0	36.7	34.2	36.6	45.0	43.5

Table 6. The ablation study on different types of injection.

Local	Global	VideoMME w/o sub.				MV-Bench	Ego-Schema
		short	mid	long	all		
direct	coarse	39.0	**36.7**	34.2	**36.6**	**45.0**	**43.5**
direct	fine	39.3	35.3	34.3	36.3	44.1	43.5
coarse	coarse	**39.8**	34.8	**35.0**	36.5	44.1	42.7
coarse	fine	38.4	36.7	34.2	36.4	44.3	43.5
fine	coarse	39.1	34.7	34.2	36.0	44.2	42.7
fine	fine	38.8	34.9	34.2	36.0	43.6	42.0

the maximum amount of instruct-relevant information while it is easily suppressed in LLaVA-Video.

Open-ended benchmarks. We also compare the performance of different models on two open-ended video benchmarks, as shown in Tab. 3. We evaluate all our results via GPT-3.5-Turbo-0125. Our HICom also achieves comparable results on both benchmarks, as arriving the SOTA on ActivityNet and getting the second best on the VideoChat-GPT Bench. Our HICom with 1328 tokens is on par with Tarsier with much more training data, and performs similarly to LLaVA-OneVision with much more training data and much more visual tokens, demonstrating the effectiveness of our compression.

Generalization Ability on Video Length. Due to the design of the local-level compression, we can easily extend the lengths of sampled frames at the inference stage for our trained model with 32 frames, rather than re-training it. As we increase the number of sampled frames, the metrics of our HICom also improve across all these benchmarks. For example, the performance on short and medium videos of VideoMME grows fast from 32 frames to 128 frames at our high compressing ratio, as more useful frames are sampled. The performance on long videos of VideoMME without subtitles changes slightly, and we argue this may caused by the extremely long videos (30-60 minutes). Both 32 frames and 128 frames are too sparse for them. Our HICom can pay attention to the most relevant parts based on the sparsely sampled frames, thus changing slightly.

4.3. Ablation Studies

To save time costs, we choose VideoChat2-IT [23] 896K video data as our instruction tuning data and Qwen2.5-0.5B [48] as our LLM to conduct ablation studies, and extract 32 frames for training unless otherwise specified. This requires less than 28 hours training on 8 A800 GPUs.

Component Analysis. The key components of our HICom are local- and global- level compression. Tab. 4 shows the results of them in the status of both unconditional (*i.e.*, without condition injection) and conditional compression (*i.e.*, with condition injection). Note we use $\boldsymbol{V}_p^{t,h,w}$ as *query* of Eq. (1) in unconditional compression for direct injection. With the same number of input tokens, the unconditional local compression can achieve comparable performance with both spatial pooling (line 2) and temporal-spatial pooling (line 3). Due to the lack of temporal-spatial inductive bias and too few tokens, the global compression significantly harms the performance, and local+global performs only slightly better than local as the additional information of the global branch is limited without explicit guidance. Compared with unconditional compression, the local and global conditional compression increase by 0.8% and 0.5%, respectively. The local+global achieves the best, increasing by 1.9% on VideoMME and 1.3% average on three benchmarks, also increasing by 0.63% compared with local conditional compression. As the global branch can provide instruction-relevant information from a global sight, combining both is better. It is noteworthy that our conditional compression with 680 tokens can beat the average pooling with 2592 tokens on all benchmarks, reducing 73.8% visual tokens and increasing by 0.47% on the performance.

Compressing Ratio. We explore the effectiveness of the compressing ratio on both unconditional and conditional compression in Fig. 5. Our unconditional compression performs similarly with average pooling on three benchmarks at different compressing ratios, and the conditional compression performs much better than them. It is noteworthy that the speed of performance decreasing of the conditional compression is also lower than the unconditional compression as the compressing ratio becomes higher. This shows instruction-relevant information remains with the guidance, demonstrating the superiority. In the lower right sub-figure of Fig. 5, the compressing ratio of (4,3,3) saves 57.68% inference time on Qwen2.5-7B than (1,3,3) with only 2.06% performance loss on Qwen2.5-0.5B. Higher compressing ratios (4,9,9) and (8,9,9) can only save a little more time with much higher performance loss. Therefore, we choose (4,3,3) as our final compressing ratio.

Conditional Pre-training Stage. To evaluate the effectiveness of our proposed conditional pre-training stage, we conduct experiments on whether to use this stage or not on both unconditional and conditional compression in Tab. 5. For average pooling and our unconditional compression, the conditional pre-training stage is also effective on MVBench and EgoSchema since more data is trained, but is hard to be effective on VideoMME. For the conditional compression,

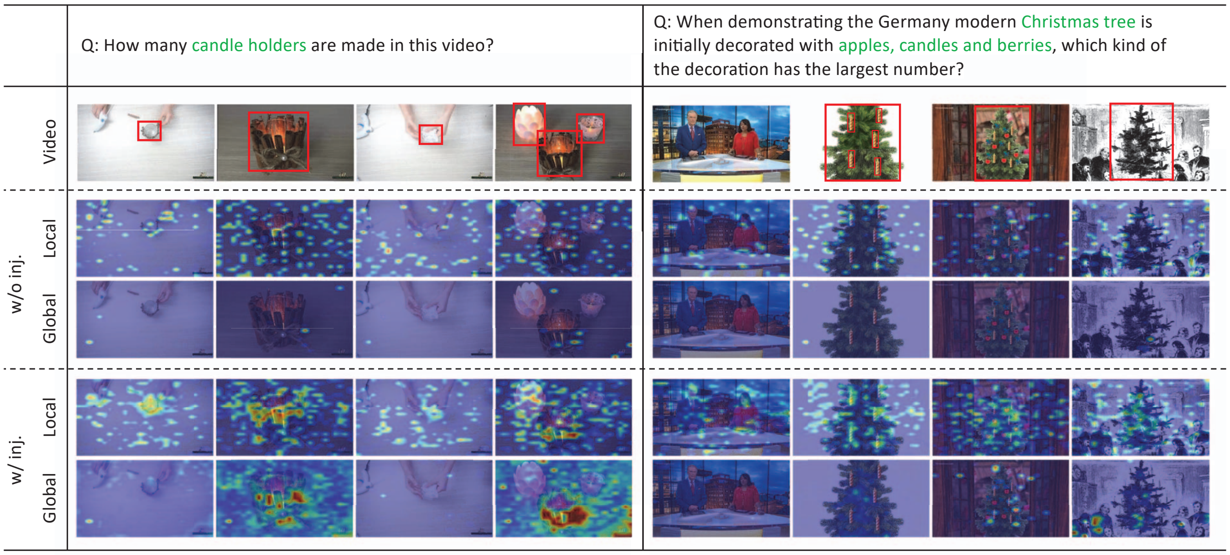

Figure 6. The visualization of the attention map at both the local and global level in the situation of unconditional (w/o inj.) and conditional (w/ inj.) compression. We indicate the instruction-relevant parts with the red bounding box for easier reading.

the conditional pre-training stage is effective on all benchmarks, increasing by 1.17% averagely, demonstrating the effectiveness of not only the constructed data, but also the newly proposed stage.

Types of Injection. We propose three types of instruction condition injection in Sec. 3.1, direct injection is designed for local compression only, while coarse and fine injection fit both. We conduct the ablation study on them in Tab. 6. Coarse injection performs better than fine injection on all three benchmarks, and both of them perform better than unconditional compression. We argue the reason may be that the pooled instruction feature is more in line with the CLIP training, and the attention of fine injection is harder to converge than the MLP of coarse injection.

4.4. Qualitative Analysis

To qualitatively figure out how our HICom realizes the conditional compression, we visualize the attention map of the compression module in Fig. 6. It clearly shows where the model pays attention to based on the instructions. Since we conduct the local-level compression within the subregion of each group, the attention map of the local level is more evenly distributed throughout the video than the global level. Compared to unconditional compression, the attention map of conditional compression is more related to the instruction-relevant visual parts. As shown in Fig. 6, the attention maps at both levels successfully focus on the candle holders in the first example, and successfully on the Christmas tree, the apples, candles, and berries in the second example, which means the model will give more weights of these parts during compressing. We also notice that the global-level compression tends to forget some detailed visual information (*e.g.*, the candle holders in the first and third frame of the first example, the Christmas tree in the second frame of the second example), while the local-level compression is better to capture these details due to group-limited attention. This shows that hybrid-level compression provides different sights, thus helping the model better understand the videos.

5. Conclusion

In this paper, we address the challenge of achieving efficient video understanding in multi-modal large language models (MLLMs) through our proposed HICom. Our approach effectively balances the trade-off between computational costs and information retention by integrating user instructions into the compression process. This method not only preserves the temporal-spatial structure of video data but also emphasizes instruction-relevant content through a dual-level compression strategy. Furthermore, by introducing a conditional pre-training stage with proposed HICom-248K dataset, targeted pre-training enhances the efficacy of instruction-injected schema. Our experiments demonstrate the effectiveness, as HICom can maintain distinguished video understanding ability with much fewer tokens. HICom shows a good generalization ability to extend the number of frames during inference, but the fixed frame sampling strategy during training still limits the ability to understand long videos. In the future, we will try to sample frames based on fps and add more long videos for training to further increase the ability. We will also explore the potential of conditional compression on images, especially high-resolution images, making our HICom more powerful.

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